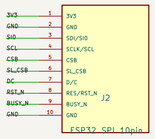
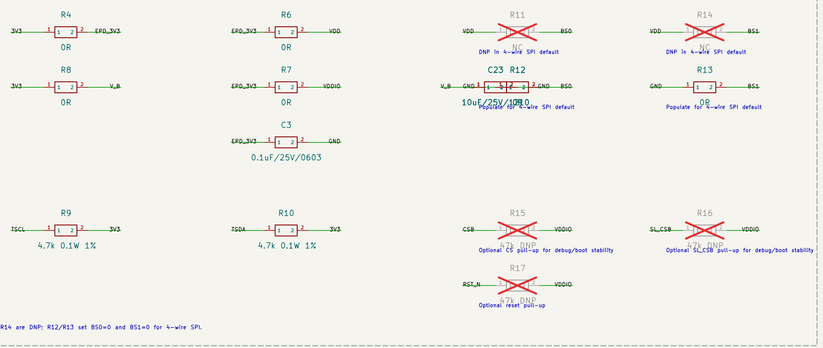
MCU / SPI control input



3V3 distribution, pullups, and 4-wire mode

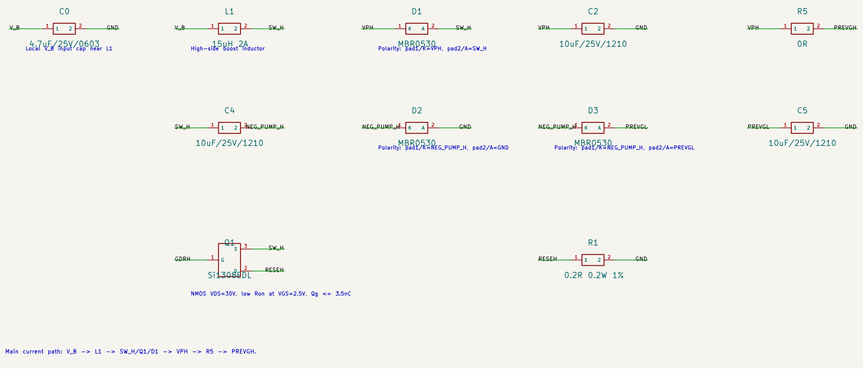


R11/R14 are DNP; R12/R13 set BS0=0 and BS1=0 for 4-wire SPI.

Panel FPC and MF-1750 signals

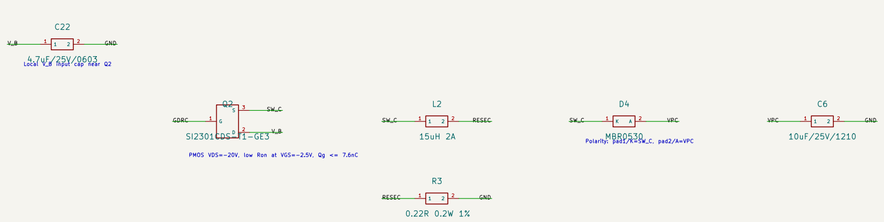


HV boost and negative gate pump



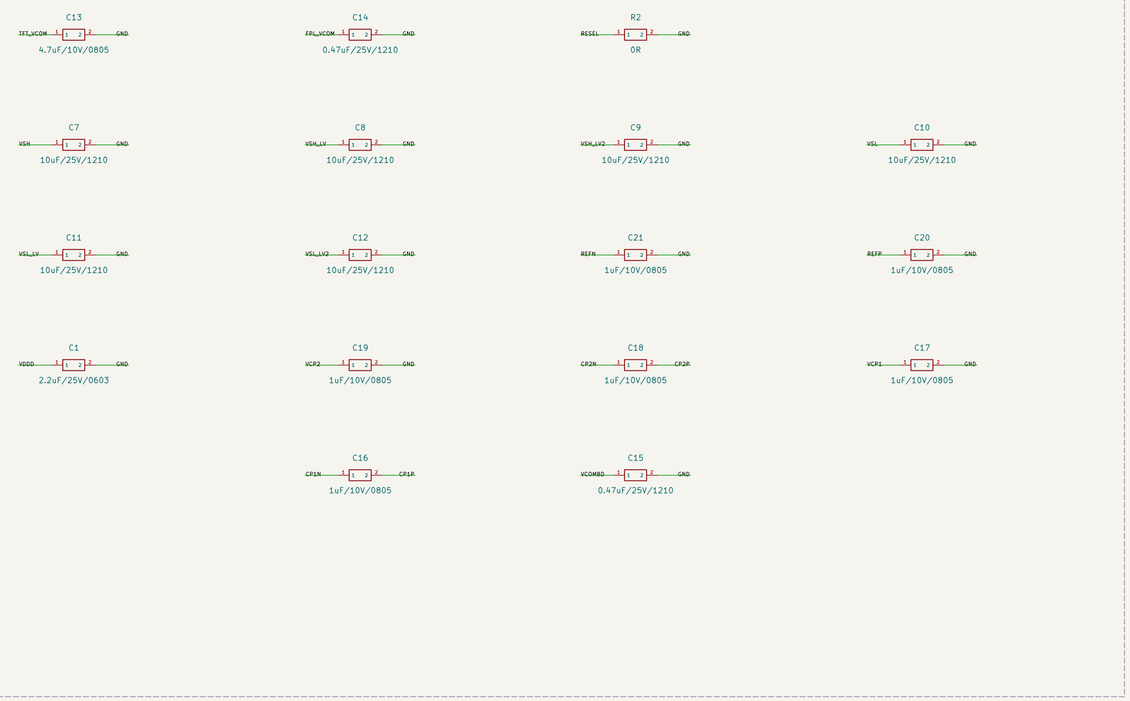
Main current path: V_B -> L1 -> SW_H/Q1/D1 -> VPH -> R5 -> PREVGH.

VPC positive pump

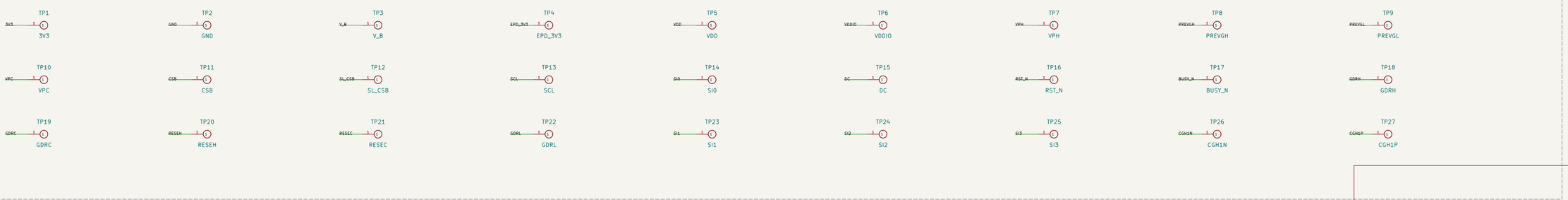


Q2/L2/D4/R3 form the VPC pump; keep this loop compact in PCB layout.

Panel-side caps, references, and flying capacitors



Bring-up and reserved test pads



Reference source: MF-1750/M009F-T specification page 10 reference circuit.

Critical fix versus DESPI-C73: J1 pin 4 is SL_CSB and must route to an ESP32 GPIO as CS2.

Functional blocks now group and wire local nodes; net labels are reserved for cross-block interfaces.

Default bus mode: BS0=0 and BS1=0 through R12/R13 for 4-wire SPI with D/C.

Diode polarity is assigned in BOM notes; match footprint pin A/K to those notes before PCB layout.

Reserved FPC pins are routed only to isolated single test pads; pin1 NC is explicitly no-connect.

Generated in ESP32_C73 workspace

Sheet: /

File: mf1750_full_driver.kicad_sch

Title: MF-1750 / M009F-T Full 50-pin Driver - Reference Schematic

Size: A2	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 1/1